

Wave Casting for Hybrid-Scene Interactive Holography

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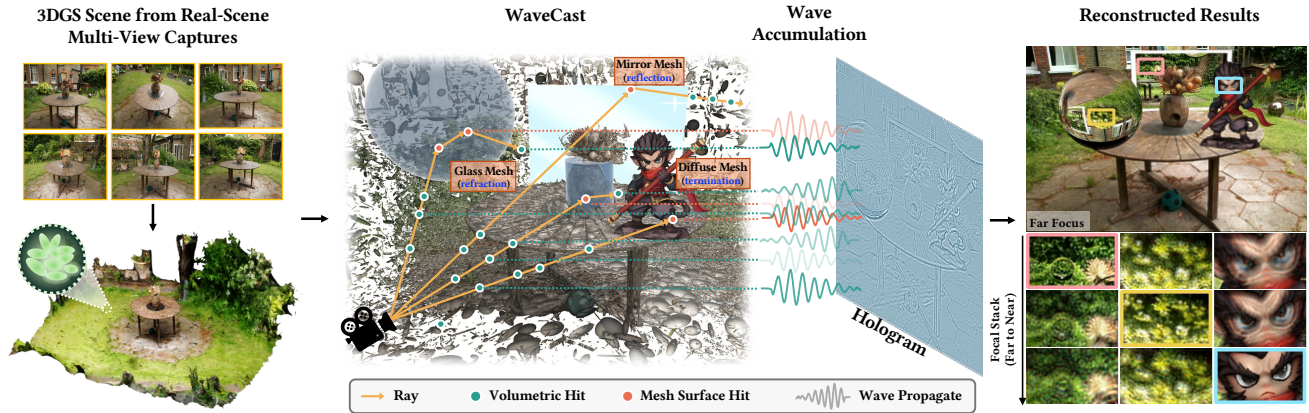


Fig. 1. Given a hybrid scene composed of a real-world 3DGS reconstruction and inserted mesh-based virtual assets, WaveCast synthesizes holograms through a coupled ray-wave process. Scene-level ray traversal identifies path interactions with volumetric primitives and mesh surfaces and accumulates them as localized subhologram contributions on the SLM plane. The resulting reconstructions show refocusable mixed-reality content with scene-dependent occlusion, reflection, and refraction at different focal depths.

Computer-generated holography (CGH) seeks to synthesize a complex wavefront on a spatial light modulator (SLM) whose free-space propagation reconstructs a target 3D scene. Existing CGH pipelines are tightly coupled to specific scene representations and provide limited support for geometry-aware light transport effects across complex 3D scenes. We introduce WaveCast, a unified ray-wave framework for hologram generation across heterogeneous volumetric and surface-based primitives, including 3D Gaussians and triangle meshes. At the core of WaveCast is *wave casting*, a ray-based process that traverses mixed geometry while accumulating transport-aware wave contributions associated with visibility, transmittance, reflection and refraction. These light-scene interactions are then converted into localized subhologram contributions that are coherently accumulated on the SLM. To enable scalable hologram synthesis, we develop a custom CUDA kernel for tiled gather-style subhologram accumulation, achieving interactive performance. Experiments on real and hybrid scenes demonstrate that, to the best of our knowledge, WaveCast is the first holographic rendering pipeline supporting hybrid real-virtual scenes and path-dependent optical effects. Our work establishes a foundation for scalable, geometry-aware holographic rendering where volumetric and surface-based scene representations can be jointly traversed and synthesized into coherent holograms.

CCS Concepts: • **Computing methodologies** → **Computer graphics**; • **Hardware** → **Emerging technologies**.

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This work is licensed under a Creative Commons Attribution 4.0 International License. SA Conference Papers '26, Kuala Lumpur, Malaysia
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ACM ISBN 979-8-4007-2842-6/2026/12
<https://doi.org/10.1145/3829340.3842287>

Additional Key Words and Phrases: computational displays, holography, mixed reality, 3D Gaussian splatting, hybrid scene rendering

ACM Reference Format:

Yujie Wang, Xi Peng, and Praneeth Chakravarthula. 2026. Wave Casting for Hybrid-Scene Interactive Holography. In *SIGGRAPH Asia 2026 Conference Papers (SA Conference Papers '26)*, December 01–04, 2026, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3829340.3842287>

1 Introduction

Holography reconstructs the optical wavefront of a 3D scene, enabling depth-dependent focus cues that are difficult to reproduce with conventional stereoscopic or light-field displays. Computer-generated holography (CGH) therefore synthesizes the field on the spatial light modulator (SLM) from scene geometry, visibility, and light transport in the 3D scene. For many years, however, efficient and high-fidelity 3D scene acquisition, representation, and rendering were not readily available. As a result, most practical CGH pipelines relied on image- or layer-based inputs, where hologram synthesis operates on 2D or 2.5D scene representations.

Recent advances in neural rendering [Kerbl et al. 2023; Mildenhall et al. 2020], particularly 3DGS [Kerbl et al. 2023], have made high-fidelity reconstruction of real scenes practical. On the other hand, authored and AI-generated virtual content remains predominantly mesh-based, including reflective and refractive objects such as mirrors and glass. A practical mixed-reality holographic pipeline should therefore support *heterogeneous 3D scenes*, where neural volumetric reconstructions coexist and interact with authored mesh geometry together with optical interactions such as occlusion, reflection,

and refraction. Existing CGH pipelines are not designed for this emerging scene format. Image- and layer-based methods reduce the scene to 2D representations early in the pipeline, limiting access to the underlying 3D geometry and visibility information. Methods based on single scene representation [Choi et al. 2025; Fricke et al. 2024; Maimone et al. 2017; Zhang et al. 2022] do not naturally combine 3DGS with meshes or propagate ray-path information into wavefront synthesis. Prior work has explored ray-traced CGH with path-dependent effects, primarily in synthetic mesh-based scenes [Blinder et al. 2021; Zhong et al. 2022], and hybrid geometric representations for efficient occlusion processing [Gilles et al. 2016]. However, current pipelines lack a unified mechanism for converting optically interacting hybrid real-virtual scenes, comprising volumetric primitives and polygon meshes, into coherent holographic wavefronts at interactive rates.

We present WaveCast, a unified ray-wave framework that couples ray tracing with coherent wavefront accumulation to generate holograms directly from hybrid volumetric and mesh scenes. The core of our approach is *wave casting*, in which rays traced through the scene determine ordered ray-primitive interactions, including volumetric transmittance and surface reflection or refraction. Instead of reducing these interactions to primitive positions alone, WaveCast propagates path-level information (e.g., accumulated transmittance, interaction order, and reflective or refractive ray redirection), into the wave accumulation stage. Each interaction is then mapped to a localized wave emitter and accumulated over a diffraction-limited subhologram footprint on the SLM plane. By preserving ray-path information during wavefront synthesis, the same pipeline can represent effects that existing primitive-position formulations cannot directly model, including reflection, refraction, and multi-bounce focus behavior.

To make this formulation practical at HD resolutions, WaveCast introduces two acceleration strategies. A *segment-level virtual source* approximates multiple volumetric interactions within a ray segment using a single equivalent emitter, reducing the number of wave contributors while preserving dominant transmittance and depth structure. A custom *tile-based gather-style CUDA kernel* reorganizes subhologram accumulation around SLM tiles, avoiding atomic-write contention from many overlapping subhologram footprints. Together, these components enable interactive hologram synthesis for real captured scenes with authored diffuse, reflective, and refractive objects.

Our implementation generates HD-resolution color holograms at 10 fps on hybrid captures and authored scenes. The resulting numerical and optical reconstructions exhibit refocusable scene content whose focus-defocus behavior follows the underlying ray-traced transport, including occlusion, reflection and refraction, and associated depth changes. Although our method is not a physically complete model for every material or optical effect, the proposed ray-wave architecture maintains path-level scene information throughout wavefront synthesis and can therefore accommodate future extensions such as continuous optical-path integration, spectral material response, and caustics within the same computational framework. We believe this ray-wave formulation provides a practical foundation for scene-aware CGH operating directly on

heterogeneous 3D scenes. Project materials and source code are available in our GitHub repository¹.

2 Related Work

Recent advances in holographic displays [Chang et al. 2020; Javidi et al. 2021; Park 2017] renewed interest in practical CGH pipelines. While our solution departs from existing approaches and pose CGH as a rendering problem, we discuss the most relevant literature here.

Primitive-based Methods. CGH approaches can be categorized by the geometric primitives used to represent 3D scenes. *Point-based methods* [Chen and Wilkinson 2009; Fricke et al. 2024; Maimone et al. 2017; Nishitsuji et al. 2021] decompose objects into discrete spherical wave emitters, offering geometric versatility. *Polygon-based methods* [Ahrenberg et al. 2008; Kim et al. 2008; Matsushima and Nakahara 2009; Zhang et al. 2022] treat objects as area sources, providing explicit surface connectivity and accurate geometric boundaries, making them well-suited for human authored assets. In practice, however, both representations often require dense sampling to faithfully represent continuous objects, incurring substantial computational cost, with undersampling the scene leading to visible artifacts. Neither photorealistically represents complex real-world scenes with intricate textures or soft volumetric effects. Moreover, since classical CGH pipelines are usually built around a particular primitive type, inserting heterogeneous content yields ambiguous occlusion ordering, making them ill-suited for hybrid scene editing.

Image-based and Learning-based Methods. Image-based CGH methods commonly synthesize holograms from representations such as multiplane images, focal stacks, or single RGB-D views. These simplify hologram generation and can produce high-quality reconstructions [Chakravarthula et al. 2019, 2020; Choi et al. 2021; Kavaklı et al. 2023; Peng et al. 2020], but they operate on view-dependent image samples rather than explicit 3D scene geometry, yielding limited viewpoint variation and weak parallax, especially under large viewpoint changes. Light-field-based approaches improve angular consistency and parallax by incorporating multi-view observations into hologram synthesis. Traditional light-field stereogram methods [Yamaguchi et al. 1993; Zhang et al. 2015] can reconstruct richer depth cues, but suffer from a spatio-angular resolution tradeoff due to the finite bandwidth of holographic displays. Optimization-based formulations using light fields [Chakravarthula et al. 2022; Choi et al. 2022; Schiffrers et al. 2023] or image-domain constraints [Chakravarthula et al. 2019; Choi et al. 2022, 2021; Kavaklı et al. 2023; Peng et al. 2020] can improve reconstruction fidelity, but incur substantial computational cost and latency due to iterative phase optimization. Recent learning-based approaches with neural networks [Choi et al. 2021; Peng et al. 2020; Shi et al. 2021; Wang et al. 2022] significantly reduce inference time; however, they are often tightly coupled to specific display hardware configurations and propagation models, and frequently require retraining when display configurations change. Both image-based and learning-based pipelines operate on static observations without an explicit scene-level rendering abstraction. As a result, these methods remain limited for dynamic and photorealistic mixed-reality holography.

¹https://github.com/rongduo/WaveCast_implementation

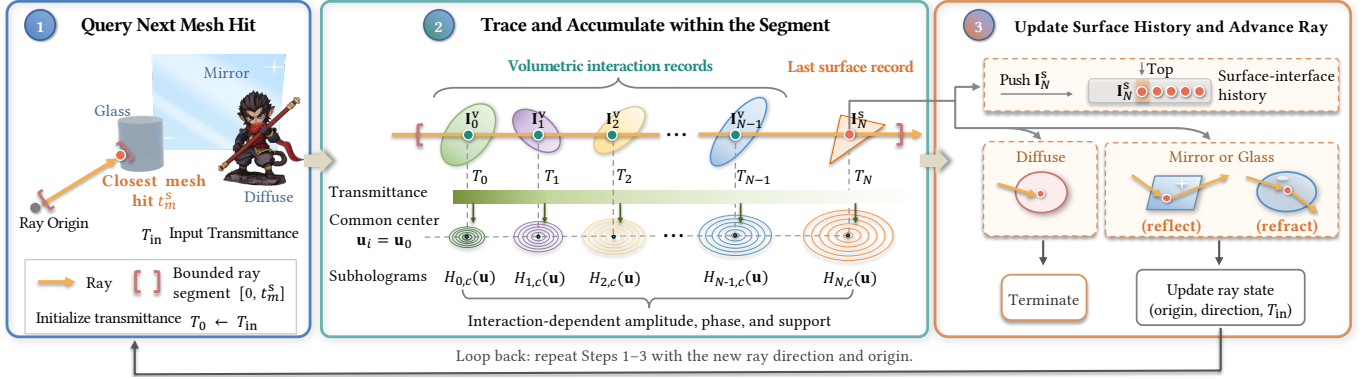


Fig. 2. **WaveCast pipeline.** For each ray emitted from an SLM pixel, we first query the nearest mesh intersection, which bounds the current ray segment. Gaussian primitives within the segment are processed in depth order together with the recorded mesh hit. Their complex subhologram contributions, all centered at the emitting SLM pixel, are weighted by front-to-back transmittance and accumulated online. After processing the terminal mesh hit, its surface interface is appended to the path history. A diffuse hit terminates the ray, whereas a reflective or refractive hit updates the ray origin and direction and initiates a new segment. This process repeats until termination.

Neural Volumetric Representations and CGH. Neural scene representations have evolved from implicit radiance fields [Mildenhall et al. 2020] and hybrid structures [Fridovich-Keil et al. 2022; Müller et al. 2022] to explicit volumetric primitives such as 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023]. While 3DGS bridges photo-realism with primitive-based flexibility, its splatting-centric rasterization is less compatible with hybrid scenes involving meshes and does not naturally support the explicit visibility reasoning required by ray-based effects, motivating recent ray-tracing extensions such as 3DGRT [Moenne-Loccoz et al. 2024]. Importantly, these representations and extensions are primarily designed for *radiance* rendering, where the goal is to reproduce view-dependent appearance. In contrast, hologram synthesis aims to generate a *coherent wavefront* on the SLM, requiring phase-consistent complex-valued accumulation and diffraction-based propagation modeling. Recent works such as Gaussian Wave Splatting (GWS) [Choi et al. 2025] and optimization-based multi-plane formulations [Zhan et al. 2025] have investigated CGH synthesis from volumetric primitives. However, these pipelines remain monolithic, inheriting the constraints of splatting-centric rendering and offering limited support for geometry-aware effects or hybrid scene editing. In contrast, WaveCast targets *interactive hologram synthesis for hybrid real-virtual scenes*, combining 3D Gaussian primitives and polygon meshes. To the best of our knowledge, it is the first CGH framework to jointly support these heterogeneous representations and their path-dependent optical interactions within a unified rendering pipeline.

3 Method

Here, we describe wave casting (see Figure 2), in which rays traverse the scene to collect transport-weighted surface and volumetric interactions along each ray path and map them to localized wavefront contributions accumulated on the SLM plane. We first describe the hybrid ray-wave transport formulation in Section 3.1. We then describe two execution granularities: a per-hit mode that

directly accumulates wavefront contributions from every encountered interaction (Section 3.2), and a segment-level abstraction that aggregates volumetric interactions along ray segments into compact virtual sources (Section 3.3). Finally, in Section 3.4, we present a tiled gather-style CUDA kernel that accelerates hologram rendering for heterogeneous scenes to interactive rates.

3.1 Hybrid Ray-Wave Transport

3.1.1 Wave Casting Over Heterogeneous Interactions. The first component of WaveCast is a coupled ray-wave rendering formulation for heterogeneous 3D scenes. We represent the scene as a collection of volumetric and surface primitives: $\mathcal{P} = \mathcal{P}^v \cup \mathcal{P}^s$, where $\mathcal{P}^v$ denotes volumetric primitives, instantiated as 3D Gaussian splats, and $\mathcal{P}^s$ denotes explicit surface primitives, instantiated as triangle meshes. Given a target viewpoint and holographic display configuration, we cast rays through $\mathcal{P}$ and index the recorded ray-scene interactions by $\mathcal{I} = \mathcal{I}^v \cup \mathcal{I}^s$, where $\mathcal{I}^v$ and $\mathcal{I}^s$ are the index sets of volumetric and surface interactions, respectively. For each $i \in \mathcal{I}$, $\mathbf{I}_i$ denotes the corresponding record, containing the position, ray distance, and transport and appearance attributes. Surface records additionally store the optical-interface information. When needed, $\mathbf{I}_i^v$ and $\mathbf{I}_i^s$ explicitly distinguish volumetric and surface records.

As Figure 2 shows, WaveCast advances each camera ray through a sequence of surface-bounded segments. For the current ray state, we first locate and cache the nearest surface hit at distance t_m^s , which bounds the m -th ray segment. We denote the volumetric-interaction indices within this segment by $\mathcal{S}_m = \{i \in \mathcal{I}^v \mid t_i^v < t_m^s\}$. Processed in front-to-back order, the corresponding interactions update transmittance and contribute opacity-weighted color. For each interaction $\mathbf{I}_i^v$ indexed by $i \in \mathcal{S}_m$, let α_i be its opacity and $\mathbf{C}_i(\mathbf{d}) \in \mathbb{R}^3$ its view-dependent RGB color along ray direction $\mathbf{d}$. Its transmittance and compositing weight are

$$T_i = T_{\text{in}} \prod_{k < i} (1 - \alpha_k), \quad w_i = T_i \alpha_i, \quad (1)$$

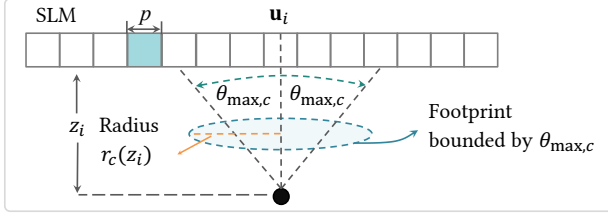


Fig. 3. **Pixel-pitch-determined diffraction footprint.** For an interaction point at depth z_i , the SLM pixel pitch p imposes a maximum diffraction angle $\theta_{\max,c}$ for wavelength λ_c , defining a footprint centered at the pixel $\mathbf{u}_i$.

where T_{in} denotes the transmittance entering the current segment. The corresponding weighted color contribution is $w_i C_i(\mathbf{d})$, with $C_{i,c}(\mathbf{d})$ denoting its value in color channel c .

After processing the volumetric interactions, the cached surface hit is recorded as a surface interaction weighted by the remaining transmittance. The surface material then determines the next ray state: diffuse surfaces terminate the path, while reflective or refractive surfaces spawn a secondary ray according to the corresponding optical-interface response. Traversal terminates when the ray hits a diffuse surface, exits the scene bounds, or its transmittance becomes negligible. This yields transport-aware records indexed by $\mathcal{I}$, where volumetric records capture soft visibility and accumulated transmittance, and surface records capture hard boundaries and reflection/refraction events. Moreover, the resulting ray segments serve as the units of the segment-level abstraction in Section 3.3.

As illustrated in Figure 2 (Step 2), each recorded interaction is treated as a wave-emitting source, often referred to below as an *emitter*, for hologram synthesis. While ray traversal determines which emitters contribute and with what weights, wave synthesis coherently accumulates their complex fields on the SLM. This ray-wave coupling forms the core of WaveCast. We next derive the mapping from scene interactions to SLM wave accumulation, starting from spherical-wave emitters and then imposing the diffraction limits of a band-limited SLM.

3.1.2 From Global Spherical Waves to Diffraction-Limited Footprints. Ideally, each interaction point $\mathbf{p}_i$, $i \in \mathcal{I}$, emits a spherical wave that contributes to all SLM pixels. For light with wavelength λ_c , the direct contribution from $\mathbf{p}_i$ to SLM pixel $\mathbf{u}$ can be written as

$$U_{i,c}(\mathbf{u}) = \frac{a_{i,c}}{\rho_i(\mathbf{u})} \exp[i(k_c \rho_i(\mathbf{u}) + \phi_{i,c})], \quad k_c = \frac{2\pi}{\lambda_c}, \quad (2)$$

where $\rho_i(\mathbf{u}) = \|\mathbf{x}(\mathbf{u}) - \mathbf{p}_i\|$ is the propagation distance, $\mathbf{x}(\mathbf{u})$ is the physical position of SLM pixel $\mathbf{u}$, $\phi_{i,c}$ is the assigned initial phase, and $i = \sqrt{-1}$ is the imaginary unit. The amplitude $a_{i,c} = w_i C_{i,c}(\mathbf{d})$ encodes the interaction's radiometric appearance and transport weight. Evaluating Equation (2) globally for all interactions on all SLM pixels is computationally prohibitive. Importantly, a discrete SLM cannot reproduce arbitrary propagation angles due to the diffraction limit imposed by its finite pixel pitch p [Maimone et al. 2017], as illustrated in Figure 3. The maximum diffraction angle is

$$\theta_{\max,c} = \sin^{-1}\left(\frac{\lambda_c}{2p}\right), \quad (3)$$

which bounds the spatial extent over which an interaction can contribute on the SLM plane. For an interaction reconstructed at depth z_i , the diffraction-limited footprint radius in pixel units is

$$r_c(z_i) = \frac{|z_i| \tan \theta_{\max,c}}{p}. \quad (4)$$

We refer to this finite footprint as the interaction's *subhologram*.

The depth z_i in Equation (4) denotes the reconstruction depth of interaction i in the holographic scene. For directly visible points, it is obtained by mapping the geometric scene depth into the target reconstruction volume. For reflected or refracted paths, however, z_i should use the *effective image depth* induced by preceding optical interfaces, rather than the raw intersection depth. For instance, a point seen in a planar mirror reconstructs at its virtual image position behind the mirror, while one viewed through a refractive slab appears at a shifted axial depth. We compute $\tilde{z}_i$ by recursively updating the effective depth from the cached optical-interface records preceding interaction i , including each interface's position, type, refractive index, surface normal, and curvature. The resulting geometric or effective depth is then mapped into the display reconstruction volume before evaluating the subhologram radius and phase. Further details are provided in the Supplementary Material.

3.2 Per-Hit Subhologram Accumulation

As described in Section 3.1.2, each emitter contributes only to the SLM pixels within its diffraction-limited subhologram footprint. Let $\mathbf{u}_i$ be the ray-origin pixel position on the SLM for interaction i , and define its local SLM-plane offset $\Delta \mathbf{u}_i = \mathbf{u} - \mathbf{u}_i$. Using the Fresnel approximation, the finite-footprint contribution of interaction i is

$$H_{i,c}(\mathbf{u}) = \mathbb{I}(\|\Delta \mathbf{u}_i\| \leq r_c(z_i)) \frac{a_{i,c}}{\gamma_c(z_i)} \exp\left[i \frac{\pi p^2 \|\Delta \mathbf{u}_i\|^2}{\lambda_c z_i}\right], \quad (5)$$

where $H_{i,c}(\mathbf{u})$ denotes the complex subhologram contribution from interaction i at SLM pixel $\mathbf{u}$, and $\mathbb{I}$ is the indicator function defining its finite spatial support. The normalization factor $\gamma_c(z_i) = \max(\sqrt{\pi r_c(z_i)^2}, 1)$ normalizes the emitted amplitude, compensating for depth-dependent changes in subhologram size. For surface interactions, $a_{i,c}$ is determined by the accumulated path throughput and the corresponding surface response.

In practice, hard subhologram boundaries via integer-rounded footprint radii can introduce visible banding artifacts. Particularly, these artifacts occur when neighboring interactions undergo small depth changes that abruptly modify their footprint size. A straightforward solution is to use a fixed subhologram footprint for all interaction points, chosen large enough to cover the maximum possible subhologram size. However, this substantially increases computational cost. Instead, we replace the hard support in Equation (5) with a smoothly tapered Tukey window [Harris 1978]:

$$H_{i,c}(\mathbf{u}) = \frac{a_{i,c}}{\gamma_c(z_i)} W_T(\ell_{i,c}(\mathbf{u}); \tau) \exp\left[i \frac{\pi p^2 \|\Delta \mathbf{u}_i\|^2}{\lambda_c z_i}\right], \quad (6)$$

where $\ell_{i,c}(\mathbf{u}) = \|\Delta \mathbf{u}_i\|/r_c(z_i)$, and $W_T(\cdot; \tau)$ is a finite-support Tukey window with roll-off parameter τ . The smooth tapering reduces boundary discontinuities while preserving compact support. This enables stable subhologram accumulation without requiring expensive per-interaction footprint refinement. Finally, the hologram is

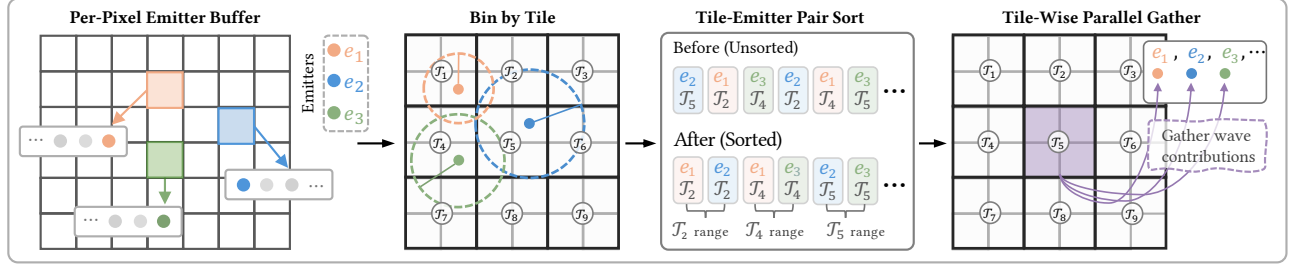


Fig. 4. **Tile-based gather accumulation.** From the per-pixel emitter buffer, we assign each emitter to the SLM tiles overlapped by its diffraction-limited footprint, forming tile-emitter pairs. After sorting by tile index, emitters for the same tile become contiguous and are loaded by a tile-wise gather kernel to accumulate complex wave contributions in shared memory. This converts wave accumulation into pixel-centric gathering, avoiding atomic-write contention.

obtained by coherent superposition of all subholograms generated by the recorded volumetric and surface interactions:

$$\mathcal{H}_c(\mathbf{u}) = \sum_{i \in I} H_{i,c}(\mathbf{u}). \quad (7)$$

The expression in Equation (7) is written in source-centric form. During parallel ray traversal, each ray records only the interactions encountered along its path. We define *scatter* as a source-centric accumulation pattern in which each wave contributor, such as a ray-scene interaction, distributes its complex wave contribution to all SLM pixels within its diffraction-limited footprint. We therefore refer to the direct per-interaction realization of Equation (7) as Mode 1: per-hit scatter. While Mode 1 most faithfully realizes Equation (7), dense scenes can generate many volumetric interactions per ray, causing large numbers of overlapping subholograms to scatter contributions into the same SLM region. This leads to substantial compute cost and memory contention. To mitigate these costs, we next introduce segment-level and tiled gather-style accumulation strategies for efficient hologram synthesis.

3.3 Segment-Level Wavefront Accumulation

Ray segments as collective rendering units. Volumetric rendering is inherently collective, which naturally motivates a coarser wavefront accumulation strategy. The appearance of a ray segment is determined by front-to-back alpha compositing of all Gaussian interactions within that segment, rather than by any single interaction in isolation. We therefore aggregate the volumetric interactions within each ray segment into a single *virtual wave source*, while retaining surface interactions explicitly. Each virtual source represents its segment with an effective depth and an aggregated amplitude, reducing the number of wave contributors while preserving segment-level appearance and transmittance.

Virtual source aggregation. Let $\mathfrak{S}$ denote the ray-segment index set. For each $m \in \mathfrak{S}$, the records $\{I_i^v\}_{i \in S_m}$ contain the volumetric interactions within that segment. Each record provides $\mathbf{p}_i$, w_i , and $C_i(\mathbf{d})$, where w_i is the transmittance-aware weight from Equation (1). All records indexed by S_m originate from the same camera ray and therefore share the lateral SLM center $\bar{\mathbf{u}}_m$. The scene-to-hologram depth mapping often compresses the depth range under the target view into a narrow reconstruction range, e.g., typically 3–10 mm in existing CGH approaches [Choi et al. 2025; Shi et al.

2021; Yang et al. 2022]. As a result, the depth variation within a local segment is also compressed into a relatively narrow range. We therefore approximate the entire segment using a single virtual source located at the transport-weighted mean depth with aggregated channel-wise amplitude:

$$\bar{z}_m = \frac{\sum_{i \in S_m} w_i z_i}{\sum_{i \in S_m} w_i}, \quad \bar{a}_{m,c} = \sum_{i \in S_m} w_i C_{i,c}(\mathbf{d}). \quad (8)$$

The subhologram is then synthesized from this single virtual source:

$$\bar{H}_{m,c}(\mathbf{u}) = \frac{\bar{a}_{m,c}}{\gamma_c(\bar{z}_m)} W_T \left(\frac{\|\Delta \bar{\mathbf{u}}_m\|}{r_c(\bar{z}_m)}; \tau \right) \exp \left[i \frac{\pi p^2 \|\Delta \bar{\mathbf{u}}_m\|^2}{\lambda_c \bar{z}_m} \right], \quad (9)$$

where $\Delta \bar{\mathbf{u}}_m = \mathbf{u} - \bar{\mathbf{u}}_m$. This replaces the exact per-hit accumulation within a segment with the compact segment-level approximation

$$\sum_{i \in S_m} H_{i,c}(\mathbf{u}) \approx \bar{H}_{m,c}(\mathbf{u}). \quad (10)$$

In practice, the quantities required for virtual-source construction are accumulated directly during traversal. Along with the total segment weight $\sum_{i \in S_m} w_i$, we maintain the weighted sums $\sum_{i \in S_m} w_i z_i$ and $\sum_{i \in S_m} w_i C_{i,c}(\mathbf{d})$, allowing the virtual-source parameters to be computed with minimal additional cost.

Note that this segment-level aggregation is applied only to volumetric interactions within each ray segment, while surface interactions are kept explicit. As a result, hard visibility boundaries, throughput updates, reflection and refraction events remain unchanged. The final hologram is therefore synthesized as:

$$\mathcal{H}_c(\mathbf{u}) = \sum_{m \in \mathfrak{S}} \bar{H}_{m,c}(\mathbf{u}) + \sum_{i \in I^s} H_{i,c}(\mathbf{u}), \quad (11)$$

where the first term sums over all ray segments and the second sums over all recorded surface interactions. We refer to this configuration as Mode 2: per-segment scatter. Compared to per-hit accumulation, Mode 2 substantially reduces the number of scattered wave emitters and therefore significantly lowers compute and memory contention.

3.4 Tiled Gather-Style Accumulation

From scatter contention to tile-based gathering. Each wave contributor emits a subhologram whose footprint grows with reconstruction depth. Even after segment-wise aggregation, neighboring contributors often produce heavily overlapping footprints on the SLM. A direct scatter implementation still triggers many concurrent

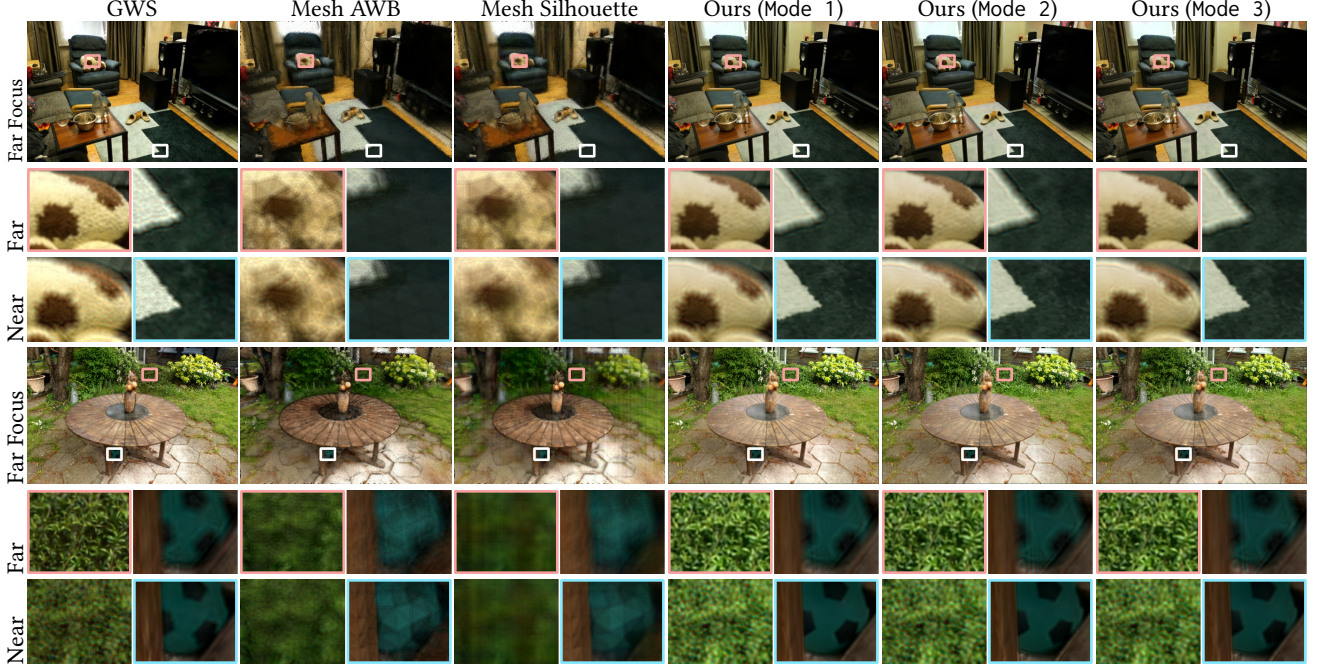


Fig. 5. **Simulated reconstructions on two single-primitive scenes.** We compare GWS and our three modes on Gaussian scenes, and Mesh AWB and Mesh Silhouette on the corresponding mesh scenes. For each scene, we show focal reconstructions at different depths and zoom-in regions. Mode 1, Mode 2, and Mode 3 correspond to different subhologram accumulation strategies, illustrating the fidelity–efficiency tradeoff in our framework.

Table 1. Summary of the three WaveCast execution modes.

Mode	Processing and accumulation
Mode 1	Per-hit emitters with scatter accumulation
Mode 2	Per-segment virtual sources with scatter accumulation
Mode 3	Per-segment virtual sources with tile-based gather

atomic additions to the same pixels, making memory contention the dominant bottleneck. We mitigate this issue by reversing the accumulation order. Rather than having each emitter scatter contributions to every pixel within its footprint, each SLM tile gathers contributions only from emitters whose footprints overlap the tile.

As shown in Figure 4, we reformulate segment-level scatter-based accumulation as tile-local gather computation. Our tiled gather pipeline proceeds in four steps. (1) *Per-pixel emitter buffering*. WaveCast stores all segment-level volumetric and surface emitters in per-pixel buffers. (2) *Binning by tile*. We partition the SLM into non-overlapping tiles $\{\mathcal{T}_t\}_{t=1}^{N_{\text{tile}}}$. Each emitter e has a subhologram support $\mathcal{D}_e$ determined by Equation (4). We conservatively bound this support using the red-channel radius, which is the largest across wavelengths. The tiles overlapped by emitter e are

$$\mathcal{A}_{\text{tile}}(e) = \{t : \mathcal{T}_t \cap \mathcal{D}_e \neq \emptyset\}. \quad (12)$$

For each $t \in \mathcal{A}_{\text{tile}}(e)$, we generate one (tile_id, emitter_idx) pair. (3) *Tile-emitter pair sorting*. We sort these pairs by tile_id, making the emitters affecting each tile contiguous in memory. (4) *Tile-wise parallel gathering*. Each tile gathers contributions only

from its associated emitters:

$$\mathcal{H}_c(\mathbf{u})|_{\mathbf{u} \in \mathcal{T}_t} = \sum_{e: t \in \mathcal{A}_{\text{tile}}(e)} H_{e,c}(\mathbf{u}). \quad (13)$$

Finally, CUDA blocks process tiles independently. Within each block, threads are assigned to output pixels in the tile, while the tile’s associated emitter list is streamed through shared memory. Each thread evaluates Equation (6) for its assigned pixel and accumulates the resulting complex contributions before writing the final value to global memory. Surface emitters and segment-level volumetric emitters share this bin-sort-gather pipeline, producing the complete hybrid hologram in a single pass.

Acceleration principle. The benefit of Equation (13) is computational: it preserves the same coherent superposition while reorganizing accumulation from emitter-centric scatter to tile-centric gather. Instead of having many emitters atomically write to overlapping SLM pixels, each output pixel is assigned to a unique thread that gathers contributions from the emitters overlapping its tile and writes once to $\mathcal{H}_c$. This eliminates atomic-write contention, improves tile-local data reuse, and keeps per-pixel complex accumulation in registers. We refer to this execution mode as Mode 3: tile-based gather, which, together with segment-level aggregation, enables interactive hologram rendering for hybrid scenes. Table 1 summarizes how the three modes progressively reduce volumetric emitters and reorganize hologram accumulation.

Table 2. Quantitative results on hologram reconstruction quality and runtime. Quality is evaluated on depth-guided All-in-Focus (AIF) reconstructions composed from 30-plane focal stacks and reported at each scene’s training resolution (Res). Runtime is measured at a common resolution of 1536×1024 . The **best** and **second-best** results are highlighted in the **Avg** row.

Scene	Res	#Prims	GWS				Mesh AWB				Mesh Sil.				Ours			
			PSNR↑	SSIM↑	LPIPS↓	Time(s)↓	PSNR↑	SSIM↑	LPIPS↓	Time(s)↓	PSNR↑	SSIM↑	LPIPS↓	Time(s)↓	PSNR↑	SSIM↑	LPIPS↓	Time(s)↓
bicycle	1237×822	866K	20.63	0.4746	0.4310	5,377.46	16.99	0.2180	0.5680	5,478.69	17.50	0.2691	0.6484	11,810.91	21.81	0.6097	0.2895	0.09
bonsai	1559×1039	637K	24.24	0.7411	0.3004	6,594.02	19.79	0.4307	0.5991	4,592.26	20.24	0.5173	0.5613	7,129.65	26.38	0.8747	0.1847	0.09
counter	1558×1038	688K	23.17	0.7320	0.2536	7,820.44	14.90	0.3651	0.5806	4,140.65	15.07	0.4127	0.5642	8,115.40	26.51	0.8638	0.1908	0.11
garden	1297×840	771K	20.32	0.5336	0.2989	6,326.07	15.35	0.2407	0.5370	7,058.96	15.76	0.2631	0.6437	14,018.91	23.15	0.7411	0.2098	0.08
kitchen	1558×1039	574K	22.55	0.7195	0.2260	9,030.59	18.31	0.3245	0.6135	4,018.39	18.50	0.3804	0.5870	7,612.30	26.28	0.8207	0.1814	0.10
room	1557×1038	546K	24.09	0.7787	0.2668	3,485.02	16.28	0.3693	0.6545	3,259.22	16.08	0.4176	0.6408	4,994.92	28.35	0.8753	0.2272	0.09
stump	1245×825	821K	21.73	0.5113	0.3799	6,392.13	18.69	0.2752	0.5420	6,128.25	19.39	0.3109	0.5870	19,395.41	24.72	0.7564	0.2465	0.09
Avg	—	701K	22.39	0.6416	0.3081	6,432.25	17.19	0.3176	0.5850	4,953.77	17.51	0.3673	0.6046	10,439.64	25.31	0.7917	0.2186	0.09

4 Experiment

4.1 Implementation

WaveCast framework. Our framework is implemented on NVIDIA OptiX with RT-core acceleration and custom CUDA kernels, using an adapted 3DGRT ray-tracing backend [Moenne-Loccoz et al. 2024]. We also build a custom holographic display prototype and use it to optically reconstruct and validate holograms rendered by WaveCast. More details are given in the Supplementary Material.

Evaluation. We evaluate on seven real-world scenes from the Mip-NeRF 360 dataset [Barron et al. 2022]: *bicycle*, *bonsai*, *counter*, *garden*, *kitchen*, *room*, and *stump*. Following the standard COLMAP split, we designate every eighth image as a test view and uniformly sample 10 views from the resulting test set for evaluation. This sampling balances view coverage against computational cost, as each baseline requires thousands of seconds to generate a single hologram. We compare with several geometry-based CGH baselines: GWS [Choi et al. 2025] which synthesizes holograms from 2D Gaussian primitives, Mesh AWB [Choi et al. 2025] which applies analytical wave-based hologram synthesis to triangle meshes, and Mesh Sil. [Matsushima et al. 2014], a mesh-based method using silhouette to handle occlusions.

We follow the GWS evaluation protocol by extracting meshes with NeRF2Mesh [Tang et al. 2023] and use the authors’ released implementations for GWS and the mesh hologram-synthesis baselines. In addition, we use the author released 2DGS checkpoints whose primitive counts match the corresponding mesh triangle counts. For WaveCast, we use 3DGRT [Moenne-Loccoz et al. 2024] with the 3DGS-MCMC training strategy [Kheradmand et al. 2024] to prepare scene assets with matched primitive budgets. Additional details are provided in the Supplementary Material. Following GWS [Choi et al. 2025], we evaluate hologram quality on all-in-focus (AIF) reconstructions using PSNR, SSIM, and LPIPS. Each AIF image is composited from 30 reconstructions focused at uniformly spaced depths across the target volume. At each pixel, the rendered depth map identifies the corresponding focal plane, from which the reconstructed pixel value is selected. Before computing metrics, we independently brightness-match each color channel to the ground-truth image. All metrics are computed at each scene’s training resolution, while runtime is measured at a common resolution of 1536×1024 . The runtime is evaluated on an NVIDIA RTX 4090 GPU.

4.2 Hologram Rendering on Single-Primitive Scenes

Table 2 compares all methods under the same per-scene primitive budget, using triangle meshes for Mesh AWB, 2D Gaussians for GWS, and 3D Gaussians for our method. For all methods, we render holograms using RGB wavelengths of 638/520/450 nm, an SLM pixel pitch of $8 \mu\text{m}$, and a 3–8 mm reconstruction depth volume. Under this configuration, the subhologram footprints in our method range from approximately 30×30 to 80×80 pixels, averaging to about 50×50 pixels. Our method achieves the highest average PSNR and SSIM and the lowest average LPIPS, reaching 25.31, 0.7917, and 0.2186, respectively, while remaining comparable to or better than GWS across individual scenes. Table 2 compares complete real-scene CGH pipelines rather than isolating the hologram-synthesis algorithms; reconstruction quality therefore reflects both scene representation and hologram synthesis.

The visual results in Figure 5 show a consistent trend. Both GWS and our method produce photorealistic reconstructions of real-world scenes, benefiting from their Gaussian volumetric scene representations, *i.e.*, 2D Gaussians for GWS and 3D Gaussians for our method. In contrast, the mesh-based baselines often show reduced texture fidelity and loss of fine structures, reflecting the difficulty of recovering high-quality meshes from complex real scenes. This is visible, for example, in the simulated tabletop and garden scenes, where the Gaussian-based methods better preserve fine tabletop textures, foliage details, and local edge structures, while the mesh-based results appear smoother. Across our three modes, the reconstructed focal stacks remain visually close, preserving similar focus-dependent details and local high-frequency patterns. The captured results in Figure 6 further confirm that our method reproduces near/far focus changes and high-frequency patterns reliably on the physical display, such as chair boundaries, foliage textures, and zoomed edge details, whereas the mesh-based baselines lose more details.

As Table 2 shows, our method is substantially faster. At 1536×1024 , GWS and Mesh AWB require thousands of seconds per view for scenes with hundreds of thousands of primitives, whereas our method takes about 0.1 seconds. On average, this corresponds to over 8.0×10^4 speedup over Mesh AWB and over 7.0×10^4 over GWS, with comparable or better AIF quality. The GWS checkpoints selected to match the mesh triangle counts contain more primitives than the approximately 15K 2D Gaussians used in the original GWS runtime evaluation, contributing to the longer runtime here.

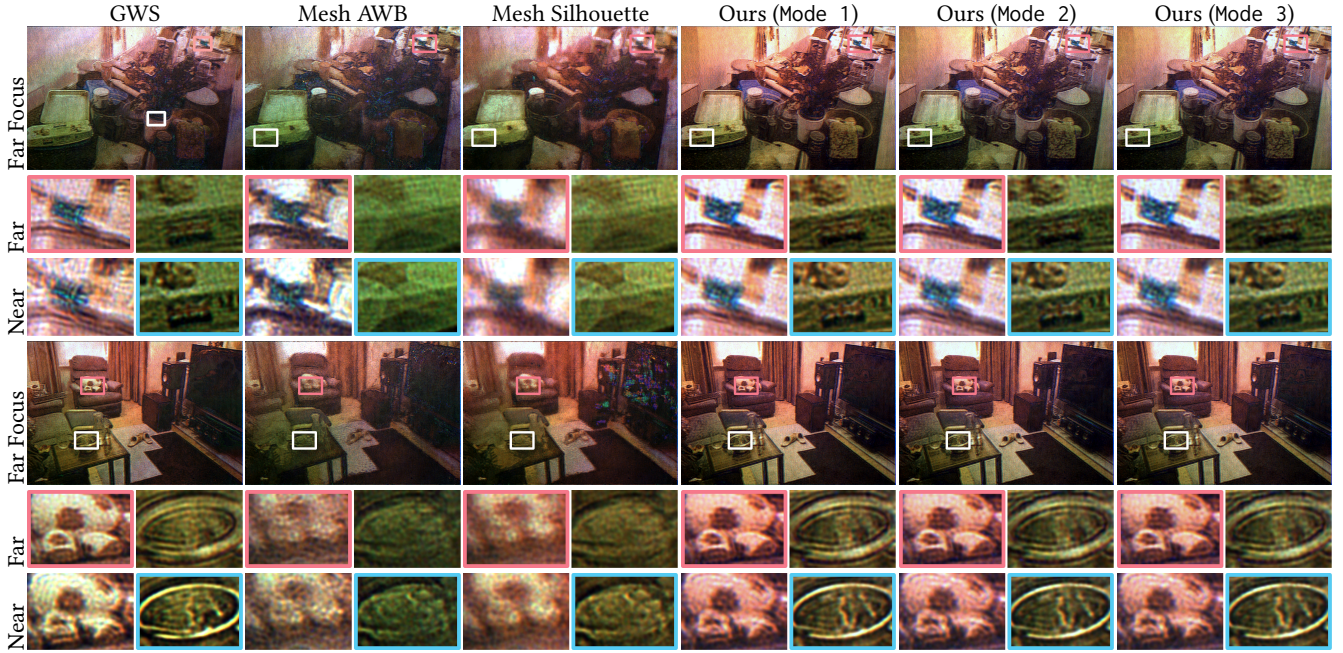


Fig. 6. **Experimentally captured results for different methods on single-primitive 3D Gaussian scenes.** For each scene, the first row shows the captured far-focus reconstruction, followed by enlarged views of the pink and cyan ROIs reconstructed at the far and near focal planes. Compared with the mesh-based baselines, GWS and our method better preserve local details across focal planes. Our three execution modes produce visually consistent in-focus details and defocus behavior despite their different accumulation strategies.

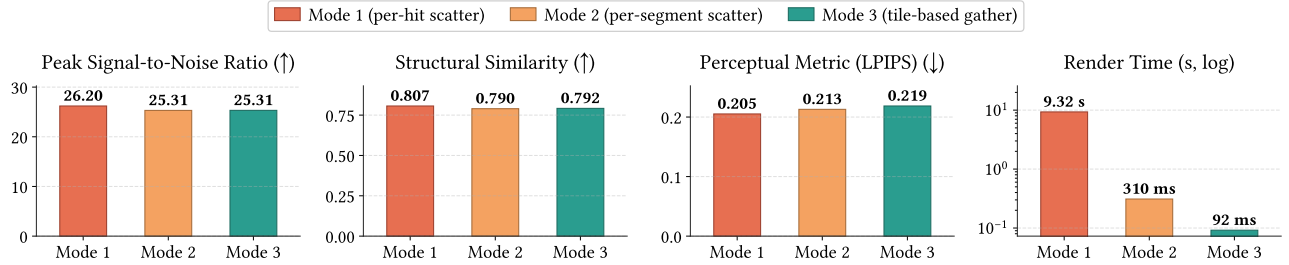


Fig. 7. **Reconstruction quality and runtime of the three modes of our method, averaged over the seven Mip-NeRF 360 scenes.** AiF Reconstruction quality is evaluated at each scene’s training resolution and the per-frame render time at 1536×1024 on a logarithmic scale. AiF reconstruction quality is nearly identical across the three modes, while the render time differs by orders of magnitude: Mode 3 (tile-based gather) is the fastest by a wide margin.

4.3 Intra-System Analysis

4.3.1 Analysis of Subhologram Accumulation Strategies. We further analyze the quality–efficiency trade-off among the three modes of our framework. As shown in Figure 7, all three modes achieve similar numerical quality under the same resolution and scene settings. Particularly, Mode 3 reorganizes the Mode 2 accumulation from *emitter-centric scatter* to *tile-centric gather* while preserving the same coherent superposition, and producing identical reconstructions. This suggests that both the segment-level abstraction and the gather-based reformulation largely preserve the reconstruction fidelity of the original per-hit formulation. The simulated and captured results in Figures 5 and 6 further support this observation. However, their

runtimes differ by orders of magnitude. The tile-based gather in Mode 3 reduces the average runtime from 9.32 s for per-hit scatter and 310 ms for per-segment scatter to 92 ms, while maintaining comparable reconstruction fidelity.

4.3.2 Runtime Scaling. Figure 8 further evaluates the runtime scaling behavior of the three modes. The first two panels sweep the number of Gaussian primitives from 20K to 1M at output resolutions of 1024×1024 and 2048×2048 . The third panel varies the holographic depth-volume span from 3 to 18 mm by fixing one depth bound at 3 mm and varying the other from 6 to 21 mm. Across all settings, Mode 1 scales steeply with scene complexity and resolution, due to the high cost of per-hit scatter accumulation over

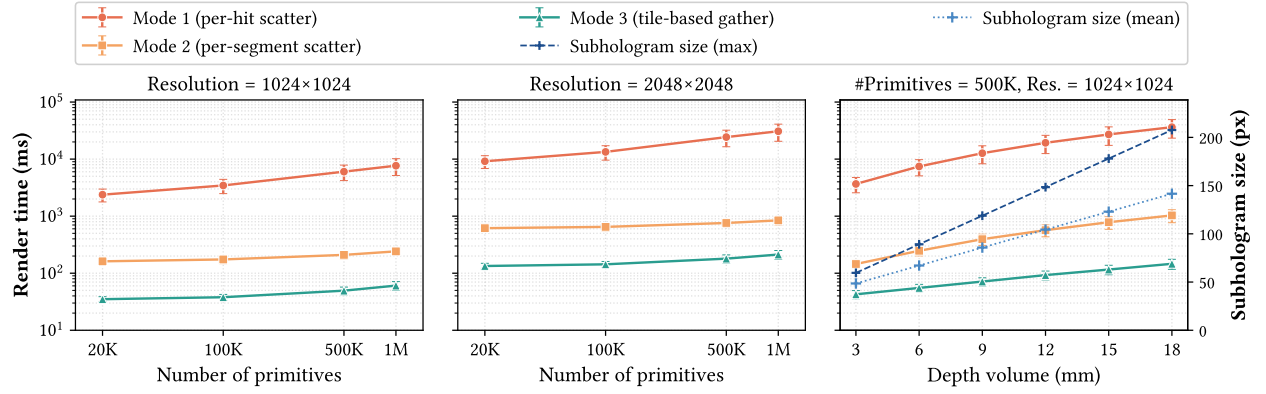


Fig. 8. **Runtime scaling of the three method variants.** We sweep Gaussian primitive count (20K–1M) at two resolutions and holographic depth volume (3–18 mm) with 500K primitives. For the depth sweep, mean and maximum sub-hologram sizes are overlaid on the secondary axis. Error bars show ± 1 standard deviation. Mode 3 remains nearly flat as complexity, resolution, and depth range increase, yielding increasing speedups over the scatter-based variants.



Fig. 9. **Captured reconstructions from our holograms for two hybrid scenes.** Across different views and focal planes, inserted diffuse objects, mirrors, and refractive spheres introduce view-dependent occlusion, reflection, and refraction effects. The results further show that reflective and refractive objects alter the effective image depth of the observed content, causing the corresponding regions to come into focus at shifted distances, as illustrated by the reflected LEGO in View 2 of the first scene and the refracted flowers in View 1 of the second scene. See more results in Supplementary Material.

many overlapping subhologram patches. Mode 2 reduces this cost through segment-level aggregation, but still performs scatter-style updates on the hologram plane. By contrast, Mode 3 consistently achieves the lowest runtime and the gentlest growth. In the depth sweep, even as the mean and maximum subhologram sizes increase with the depth volume, Mode 3 remains relatively flat compared with the scatter-based variants. This confirms that tile-based gather

accumulation scales more favorably to higher resolutions, larger primitive counts, and wider reconstruction depth volumes.

4.4 Hologram Rendering on Real-Virtual Scenes

Figure 9 shows captured focal stacks for two hybrid scenes rendered by our method. The inserted diffuse objects, mirrors, and refractive spheres are coherently integrated with the real-scene

reconstructions, introducing view-dependent occlusion, reflection, and refraction effects. Importantly, reflection and refraction change not only the local appearance but also the effective image depth of the observed content. For instance, the reflected LEGO in the first scene and the refracted flowers in the second scene are still sampled by ray casting at their actual geometric locations. However, the preceding mirror and refractive sphere remap their effective image depths, causing these regions to come into focus at shifted focal planes rather than at the original geometric depths of the sampled points. These results demonstrate that WaveCast reproduces transport-induced appearance changes and their associated effective-depth shifts in the experimentally captured hybrid scenes. The captured focal-stack sequences in the Supplementary Video further illustrate focus transitions on our display prototype.

5 Discussion and Conclusion

In this work, we introduced a unified ray-wave framework that formulates hologram synthesis as a rendering problem coupling geometric light transport with coherent wave accumulation. Our method combines ray tracing over heterogeneous scene primitives with wave-based hologram formation, enabling volumetric and surface-based content to coexist within the same holographic pipeline. This formulation allows reconstructed real scenes and inserted virtual objects to interact through scene-level visibility, occlusion, reflection, refraction, and volumetric transmittance while remaining suitable for direct optical reconstruction. Our framework enables interactive hologram synthesis for complex hybrid scenes while preserving focal cues and scene-dependent optical effects.

Several opportunities for future work exist. Although our framework correctly handles geometric transport effects for inserted virtual objects, it cannot yet reproduce fully consistent cast shadows or global illumination effects, e.g., color bleeding, between real and virtual scene components. Future integration with inverse-rendering-based scene representations could enable more physically grounded real-virtual composition with editable materials and lighting. We observe that the captured results can be affected by coherent artifacts and residual color-sequential synchronization errors, which can reduce the visual clarity of subtle focus-defocus transitions. A promising direction is to make WaveCast differentiable and incorporate learned system proxies [Chakravarthula et al. 2020; Peng et al. 2020] for end-to-end optimization of optical reconstruction quality. More broadly, our formulation opens the door to integrating modern neural scene representations, physically based rendering, and coherent wave optics within a unified computational framework.

Our work demonstrates that transport-aware holographic rendering over heterogeneous 3D scenes can be practical and scalable. Beyond holography, the proposed tiled gather accumulation strategy may also benefit broader imaging and graphics problems that require efficient accumulation of spatially varying kernels over large-scale scene representations. We hope this work helps move holography closer to a general-purpose 3D display technology capable of seamlessly bridging neural scene reconstruction, physical light transport, and interactive mixed reality.

Acknowledgments

This work has been supported by the National Science Foundation (NSF) grant #2107454.

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